

EAST WEST UNIVERSITY

Department of Computer Science and Engineering
CSE475 — Machine Learning, Summer 2026

Report 2: Baselines and Proposed GNN Model

ds004504 EEG Dataset — Alzheimer's, Frontotemporal Dementia and Healthy Controls

Dataset: OpenNeuro ds004504 – EEG in Alzheimer's Disease,
Frontotemporal Dementia and Healthy Controls

Track: 1 – Graph Neural Network on Table-Style EEG Data

Submitted by – Group 03

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Date of Submission: 28 August, 2026

This report accompanies the notebooks `task2.baselines.ipynb` and `task2.proposed_model.ipynb`.

1. Feature and Graph Construction

Per subject, raw EEG was band-pass filtered (0.5–45 Hz), split into fixed 8-second epochs, and a Welch power spectral density was computed per channel. Relative band power in five canonical bands (delta, theta, alpha, beta, gamma) was averaged over epochs, giving a 19×5 node-feature matrix per subject. Two functional-connectivity graphs were also computed and cached: Pearson correlation (broadband) and PLV (alpha band), since no single estimator is known to consistently outperform the other. Data was split subject-wise (train = 59, val = 11, test = 18), stratified by class, with scaling statistics fit only on the training subjects.

2. Task 2a – Non-Graph ML Baselines

Five classical baselines (Logistic Regression, SVM-RBF, Random Forest, XGBoost, MLP) were trained on the identical flattened 95-dimensional (19 channel $\times$ 5 band) feature vector that the GNN later uses as node features, all class-weighted to account for the mild class imbalance.

Model	Accuracy	Macro-F1	Macro-Prec.	Macro-Rec.
SVM-RBF	0.611	0.595	0.594	0.602
LogReg	0.500	0.464	0.469	0.471
MLP	0.500	0.458	0.463	0.471
XGBoost	0.444	0.383	0.357	0.413
RandomForest	0.444	0.357	0.315	0.413

Table 1: Baseline test-set results, sorted by macro-F1. SVM-RBF is best (macro-F1 = 0.595).

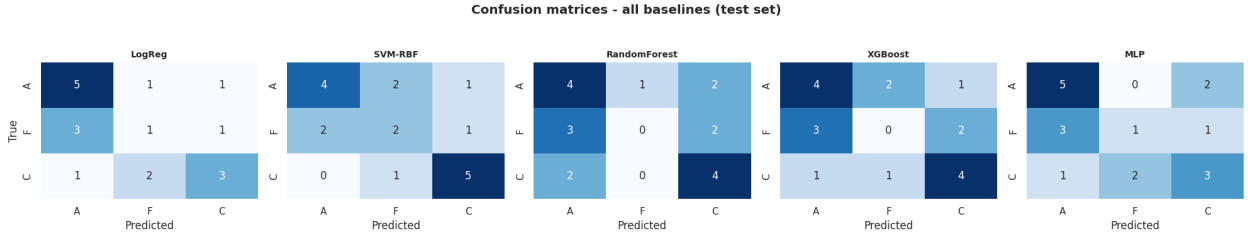


Figure 1: Confusion matrices for all five baselines on the held-out test set.

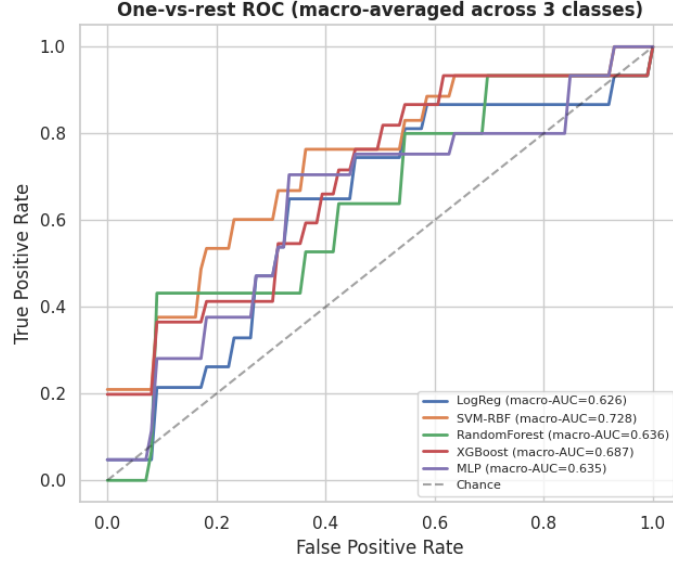


Figure 2: One-vs-rest macro-averaged ROC curves for all baselines.

3. Task 2b – Proposed GNN Model

A per-subject brain graph was built with 19 nodes (EEG channels) and edges from the subject’s own correlation-based connectivity matrix (fully connected, $|r|$ as edge weight, no self-loops). Building each graph from only that subject’s own recording ensures graph construction cannot leak across the train/test split.

Architecture

A 2-layer Graph Convolutional Network (GCN):

$$\text{GCNConv}(5 \rightarrow 32) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{GCNConv}(32 \rightarrow 32) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{global mean pool} \rightarrow \text{Linear}(32 \rightarrow 3)$$

This architecture was kept deliberately simple as a first result; Task 3 ablates depth, width, and alternative architectures.

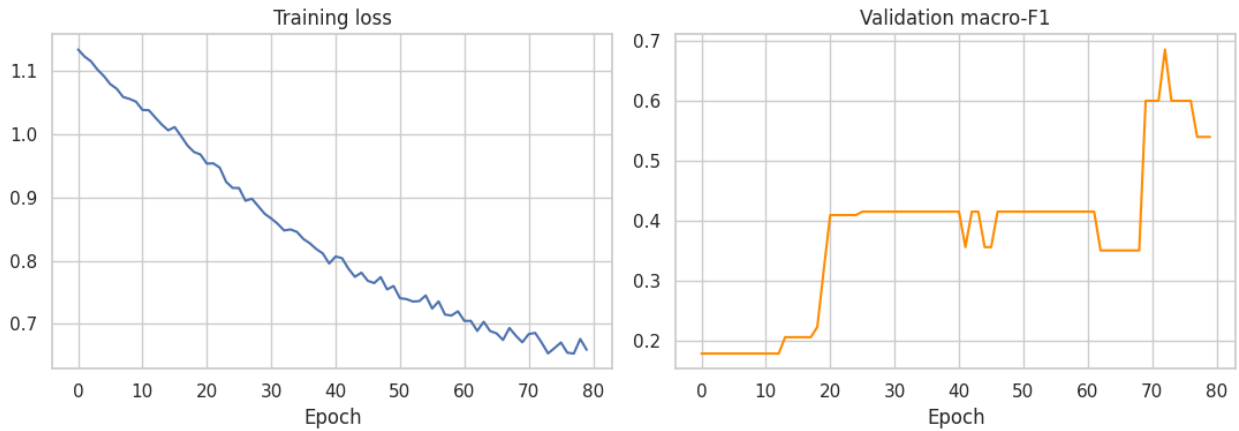


Figure 3: Training loss and validation macro-F1 across 80 epochs. Best validation macro-F1 = 0.686.

First GNN Result vs. Baselines

Model	Test Macro-F1
SVM-RBF (best baseline)	0.595
LogReg	0.464
GNN (GCN, first result)	0.458
MLP	0.458
XGBoost	0.383
RandomForest	0.357

Table 2: Test macro-F1 comparison: first GNN result vs. all Task 2a baselines.

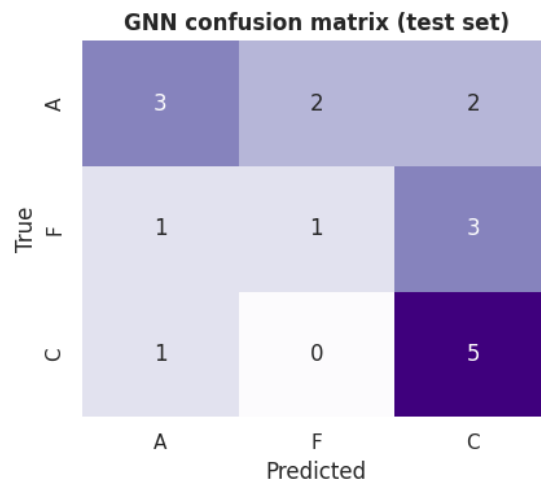


Figure 4: GNN confusion matrix on the held-out test set (accuracy = 0.500, macro-F1 = 0.458).

4. Summary

The simple first-pass GCN (test macro-F1 = 0.458) does not yet outperform the best non-graph baseline SVM-RBF (test macro-F1 = 0.595) on a single train/test split. This motivates the systematic ablation and improvement study carried out in Task 3, which sweeps graph type, architecture, depth, and width, and re-evaluates with a leakage-safe 5-fold cross-validation.